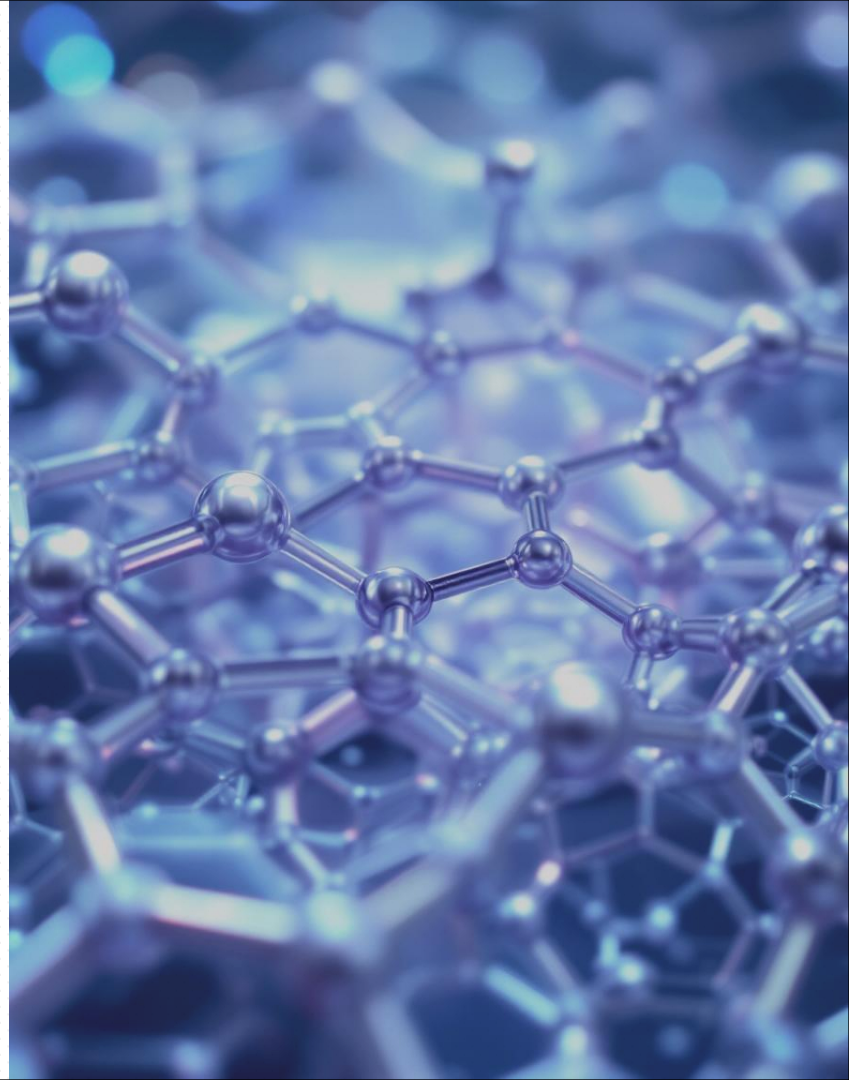


Graphene

The Technological Revolution of Future Materials

English Practice Presentation

Presented By **ZIANE Ceryne**



Outline

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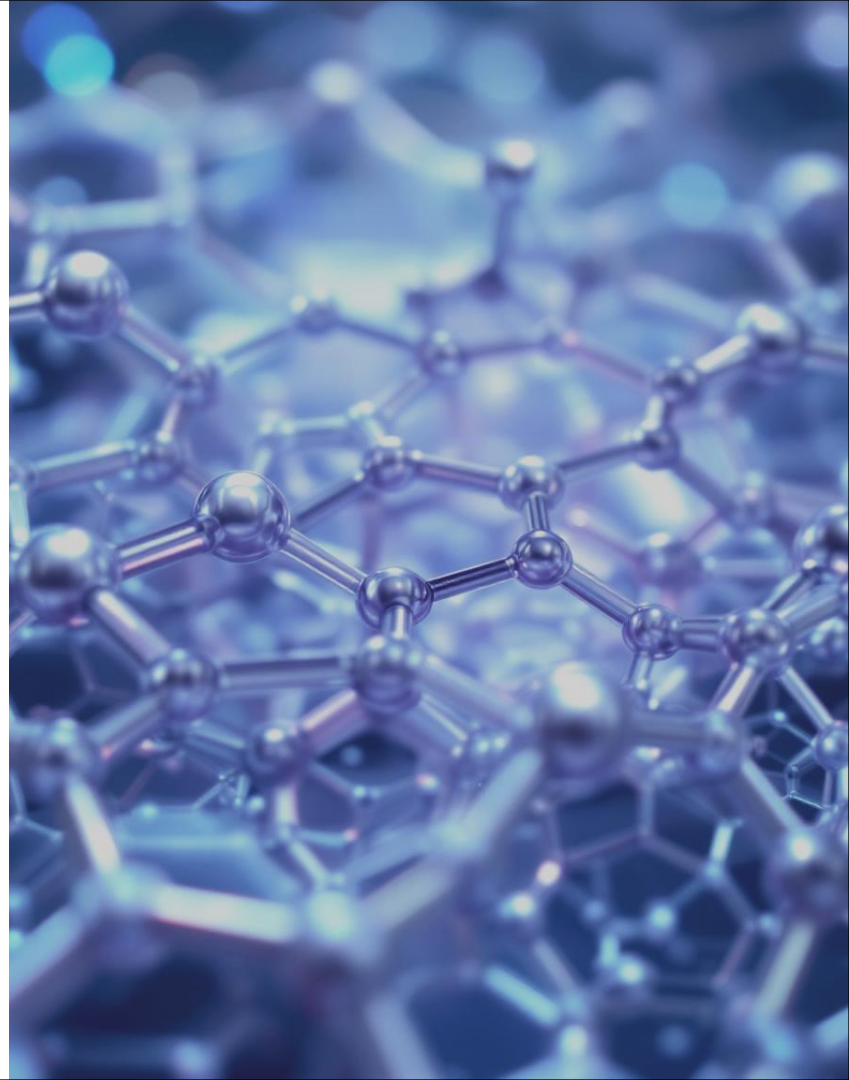
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What is Graphene? The Ultimate 2D Material

Atomic Architecture

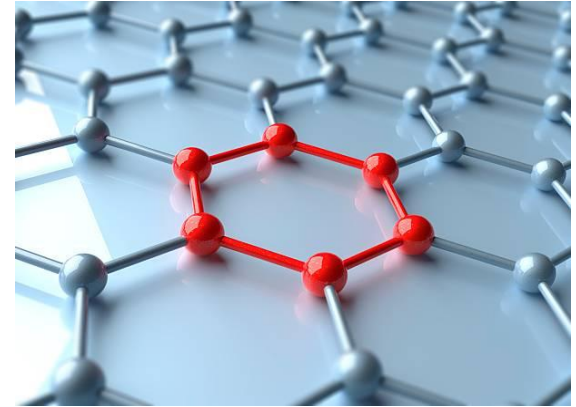
Single layer of carbon atoms arranged in a **hexagonal lattice** pattern.

Discovery : 2004 by Andre Geim and Konstantin Novoselov

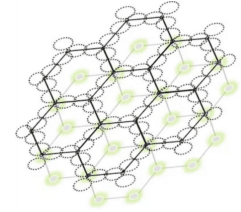
Isolated using **adhesive tape** to peel layers from **graphite** (material found in pencil lead).

Recognition : Nobel Prize in Physics, 2010.

This basic structure endows graphene with **unique properties**, earning it the title of a '**material of the future**'.



What is Graphene? The Ultimate 2D Material



Exceptional Properties

Atomically Thin

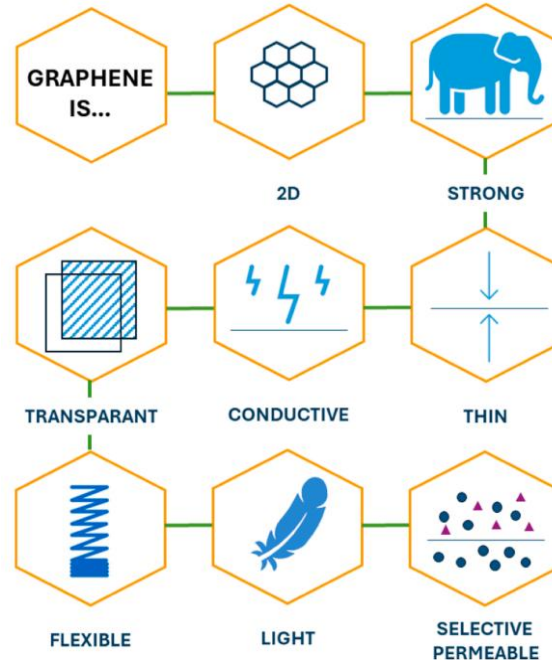
Only **one atomic layer** thick

Mechanical Strength

200x stronger than steel with exceptional **mechanical durability and flexibility**

Electrical Conductivity

High electron **mobility** and **thermal conductivity**



Heat Conductivity

Dissipates heat effectively

Optical Transparency

Transmits up to **97.7 %** of visible light

Graphene's Game-Changing Applications in IT

High-speed Transistors

Its high electron mobility allows transistors to operate at significantly **higher frequencies** than silicon, enabling **faster processors** and improved **computational performance**.

Flexible Electronics

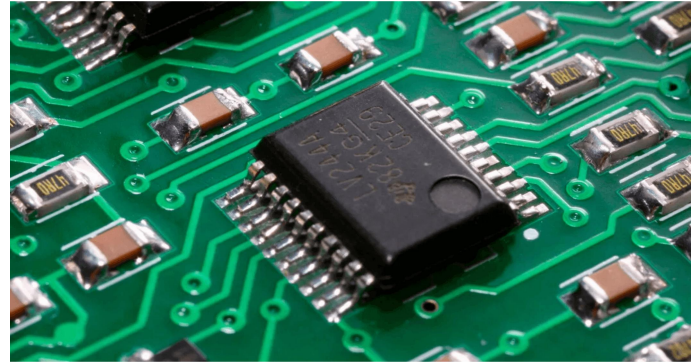
Its conductivity and transparency makes it ideal for **bendable screens and wearables**. Enabling unbreakable smartphones and **advanced flexible displays**.

Advanced batteries

Graphene supercapacitors and batteries charge very fast and hold a lot of energy. These could let electric cars charge in minutes and last much longer.

Thermal management

Lowers **power loss** in chips and sensors for more **efficient devices**.



Graphene's Game-Changing Applications

Environmental Sensors

Graphene-based sensors provide **real-time monitoring** of biological or chemical metrics with extreme **sensitivity and accuracy**. Used in **Water Desalination** and **Carbon Capture**.

Optical communication

Ultrafast photodetectors and modulators for high-speed data transmission and reception.

Lightweight composites

Graphene-reinforced materials **improve strength** by over 50%, used in **aerospace, automotive, and construction**.

Resists vibration and temperature extremes, improving **safety and lifespan** of components.

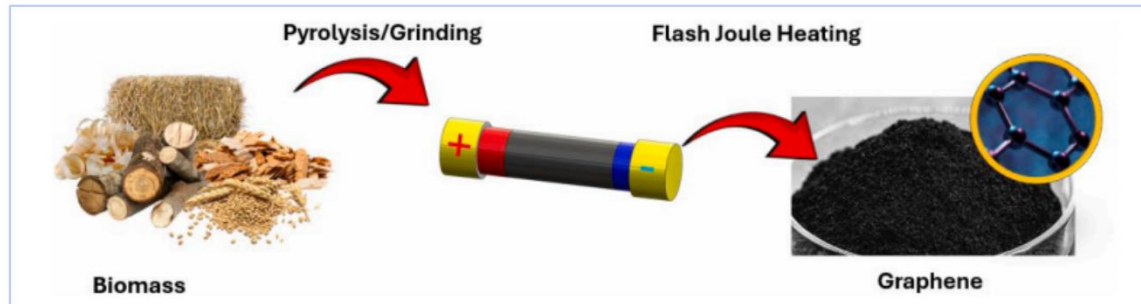


Production Challenges Holding Graphene Back

High Production Costs

Transitioning from laboratory-scale synthesis to industrial production without **compromising graphene's quality** remains a major technical challenge.

High-quality graphene requires **expensive methods** like chemical vapor deposition (CVD). Industrial grade production costs too much for mass adoption.



Production Challenges Holding Graphene Back

Quality Control Issues

Standardized testing protocols still emerging. Measurement techniques improve **reproducibility and reliability**, which are crucial for large-scale implementation.

Environmental Concerns

Some production methods release **toxic chemicals**, which may pose risks to ecosystems and human health.

Material stability and performance

Highly sensitive to environmental changes and interactions with other materials.



Emerging Opportunities

Industry 4.0 Smart Factories

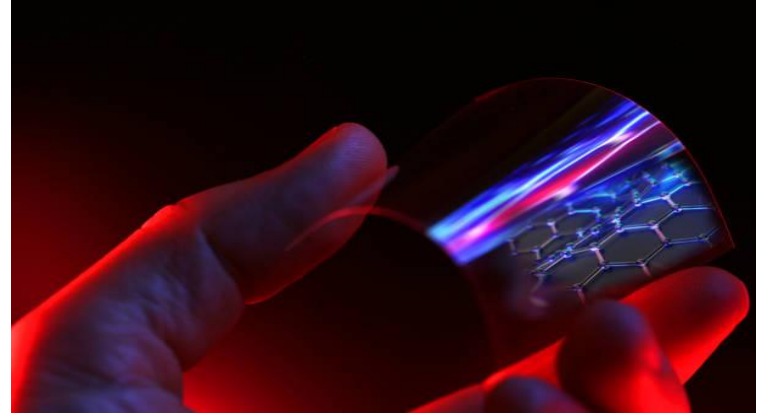
Graphene sensors enable real-time monitoring with high sensitivity and low power use. The combination of its properties makes graphene ideal for **industrial IoT applications**.

Energy-Efficient Data Storage

Graphene-based memory technologies promise faster read/write speeds and lower power consumption, that could **revolutionize data center efficiency**.

Flexible Solar Cells

Transparent, conductive graphene electrodes enable **flexible photovoltaic devices** that can be integrated into surfaces, buildings, and portable electronics, powering the next generation of **autonomous IoT devices**.



Emerging Opportunities

Quantum Computing Devices

Graphene's unique electronic properties make it a promising platform for **quantum components**.

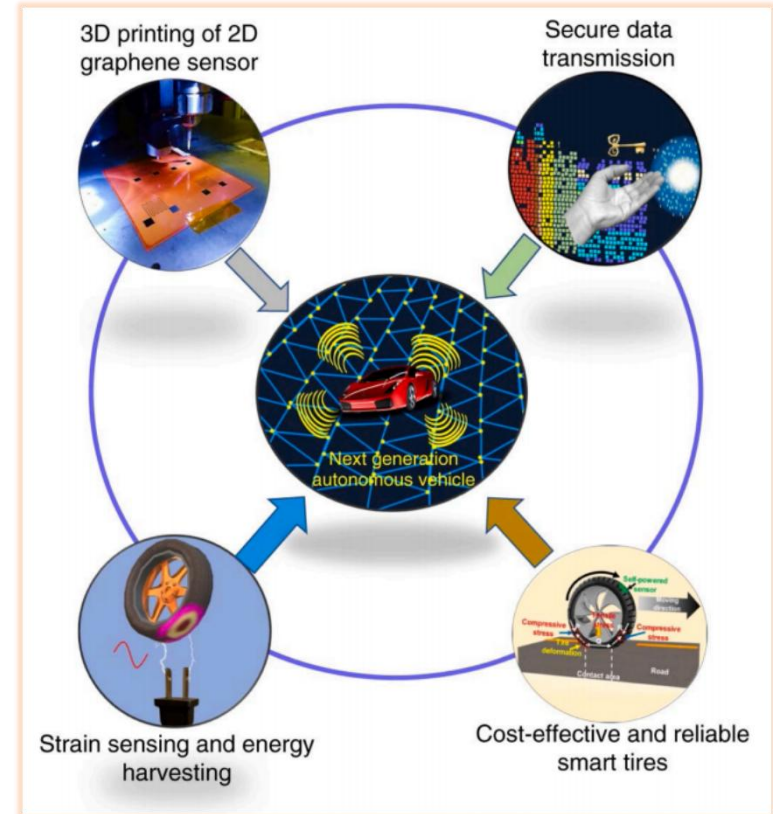
Research into graphene-based quantum devices could unlock **computational breakthroughs** beyond classical limits.

Consumer products

Using in **dresses** to using it in sports equipment, since the incorporation of graphene makes the product more efficient and resistant to wear and tear.

Hardware Security Enhancements

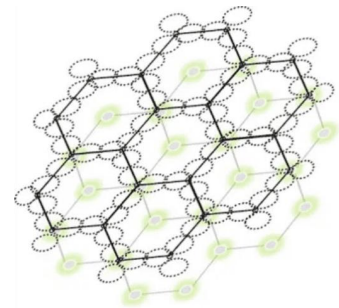
Its high electron mobility allows transistors to operate at significantly **higher frequencies** than silicon, enabling **faster processors** and improved **computational performance**.



Conclusion

Graphene's remarkable mix of **properties** makes it a game-changer for **modern technology**.

The **future of graphene** lies in its ability to bridge disciplines. Advances in production techniques, could revolutionize how we build **electronics, communications, energy devices, and sensors**.



Thank you

Questions?

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